

Shitao Jin

PhD Candidate in Architecture | AI-Enabled Architectural Programming & Megaproject Decision-Making

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PhD candidate in Architecture at Tongji University and joint-training researcher at The University of Melbourne. My work links architectural programming, construction/project management, and AI-enabled decision-making for megaprojects, transportation hubs, child-friendly communities, and learning environments.

289 Google Scholar citations	8 h-index	21 publications / papers	146 verified peer reviews	2 PhD national scholarships	7.0 IELTS
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RESEARCH PROFILE & ACADEMIC POSITIONING

- Built a coherent research program around architectural programming as the front-end decision system of complex public projects.
- Developed AI-enabled and multi-criteria decision-making approaches for megaproject programming, risk assessment, complexity measurement, stakeholder analysis, and public concern analysis.
- Extended architectural programming research from building briefing into urban community renewal, child-friendly environments, transportation hubs, and social infrastructure.
- Combined bibliometrics, topic modeling, systematic review, and empirical case analysis to map research frontiers in architectural programming and megaproject management.
- Architectural programming / briefing for complex public projects
- AI-enabled decision-making, collective intelligence, and multi-agent reinforcement learning
- Megaproject governance: complexity, risks, stakeholders, public concerns, and transportation hubs

EDUCATION & VISITING APPOINTMENT

SEP 2025 - PRESENT

Joint PhD Training / Visiting Graduate Researcher, The University of Melbourne
Faculty of Engineering and Information Technology, Department of Infrastructure Engineering. Supervisors: A/Prof Felix Kin Peng Hui; A/Prof Paulo Vaz Serra.

SEP 2023 - JUN 2027 (EXPECTED)

Ph.D. in Architecture, Tongji University
College of Architecture and Urban Planning, Department of Architecture. GPA: 3.60/4.00. Supervisor: Prof. Huijun Tu.

SEP 2020 - JUN 2023

Master of Architecture, Huazhong University of Science and Technology
GPA: 3.48/4.00.

SEP 2015 - JUN 2020

Bachelor of Architecture, Jilin Jianzhu University

SELECTED RESEARCH CONTRIBUTIONS & CURRENT AGENDA

- Built a coherent research program around architectural programming as the front-end decision system of complex public projects.
- Developed AI-enabled and multi-criteria decision-making approaches for megaproject programming, risk assessment, complexity measurement, stakeholder analysis, and public concern analysis.
- Extended architectural programming research from building briefing into urban community renewal, child-friendly environments, transportation hubs, and social infrastructure.
- Combined bibliometrics, topic modeling, systematic review, and empirical case analysis to map research frontiers in architectural programming and megaproject management.
- Current agenda focuses on mega transportation hubs, urban community renewal, and child-friendly public environments, emphasizing AI-supported collaboration among experts, publics, and stakeholders.

PUBLICATIONS & CONFERENCE PAPERS

Peer-reviewed journal articles

1. Shitao Jin. Dynamics and Influences Analysis of Public Concerns in Mega Construction Projects: An Integrated Topic and Sentiment Modeling Approach.

Journal of Construction Engineering and Management, 152(8), 04026104. DOI: [10.1061/JCEMD4.COENG-17561](https://doi.org/10.1061/JCEMD4.COENG-17561)

2. Shitao Jin; Xudong Miao. Risk Identification and Assessment of Mega Construction Projects in Uncertain Environments: A Gray Fuzzy Group Decision-Making and Monte Carlo Simulation Method.

Journal of Architectural Engineering, 32(2), 04026010. DOI: [10.1061/JAEIED.AEENG-2153](https://doi.org/10.1061/JAEIED.AEENG-2153)

3. Huijun Tu; Shitao Jin; Xudong Miao; Cong Zhou; Yuming Shi. A value-based programming framework for enhancing children's well-being in urban communities.

Frontiers of Architectural Research. DOI: [10.1016/j.foar.2026.02.012](https://doi.org/10.1016/j.foar.2026.02.012)

4. Yuan Deng; Shitao Jin; Xiaoming Zhu. Evolving public perceptions of the renewal of Suzhou neighborhood centers: An integrated sentiment and topic modeling framework.

Frontiers of Architectural Research. DOI: [10.1016/j.foar.2026.02.005](https://doi.org/10.1016/j.foar.2026.02.005)

5. Huijun Tu; Xudong Miao; Shitao Jin. A deep learning-based evaluation system for child-friendly urban streets integrating abstract and concrete features-A case of Shanghai Urban Street.

PLOS One, 21(2), e0342430. DOI: [10.1371/journal.pone.0342430](https://doi.org/10.1371/journal.pone.0342430)

6. Shitao Jin. Current status and trends of megaproject research: bibliometric and text mining analysis.

Engineering, Construction and Architectural Management, 33(6), 4543-4573. DOI: [10.1108/ECAM-04-2024-0495](https://doi.org/10.1108/ECAM-04-2024-0495)

7. Shitao Jin. Measuring complexity in mega construction projects: fuzzy comprehensive evaluation and grey relational analysis.

Engineering, Construction and Architectural Management, 33(1), 284-317. DOI: [10.1108/ECAM-07-2024-0951](https://doi.org/10.1108/ECAM-07-2024-0951)

8. Shitao Jin; Yuan Deng. Identification and evaluation of key stakeholders in mega construction projects: social network analysis and multi-criteria decision-making.

Engineering, Construction and Architectural Management, 1-38. DOI: [10.1108/ECAM-01-2025-0097](https://doi.org/10.1108/ECAM-01-2025-0097)

9. Shitao Jin. Interdisciplinary perspective on architectural programming: current status and future directions.

Engineering, Construction and Architectural Management, 32(11), 7332-7372. DOI: [10.1108/ECAM-04-2024-0408](https://doi.org/10.1108/ECAM-04-2024-0408)

10. Shitao Jin. The application of collective intelligence in the construction industry: a review of the current state, challenges, and opportunities.

Architectural Engineering and Design Management, 21(5), 1091-1116. DOI: [10.1080/17452007.2025.2478021](https://doi.org/10.1080/17452007.2025.2478021)

11. Shitao Jin; Huijun Tu. Current status and research progress in architectural programming: A comparative analysis between China and other countries.

Frontiers of Architectural Research, 14(2), 487-509. DOI: [10.1016/j.foar.2024.07.010](https://doi.org/10.1016/j.foar.2024.07.010)

12. Huijun Tu; Xudong Miao; Shitao Jin; Jiayi Yang; Xinyue Miao; Jiale Qi. Deep Learning-Based Systems for Evaluating and Enhancing Child-Friendliness of Urban Streets-A Case of Shanghai Urban Street.

Buildings, 15(13), 2291. DOI: [10.3390/buildings15132291](https://doi.org/10.3390/buildings15132291)

13. Shitao Jin; Huijun Tu; Jiangfeng Li; Yuwei Fang; Zhang Qu; Fan Xu; Kun Liu; Yiquan Lin. Enhancing Architectural Education through Artificial Intelligence: A Case Study of an AI-Assisted Architectural Programming and Design Course.

Buildings, 14(6), 1613. DOI: [10.3390/buildings14061613](https://doi.org/10.3390/buildings14061613)

14. Huijun Tu; Shitao Jin. A group decision-making model for architectural programming in megaprojects.

Engineering, Construction and Architectural Management, 31(13), 342-368. DOI: [10.1108/ECAM-03-2024-0394](https://doi.org/10.1108/ECAM-03-2024-0394)

15. Shitao Jin; Lei Peng. Classroom perception in higher education: The impact of spatial factors on student satisfaction in lecture versus active learning classrooms.

Frontiers in Psychology, 13, 941285. DOI: [10.3389/fpsyg.2022.941285](https://doi.org/10.3389/fpsyg.2022.941285)

16. Lei Peng; Shitao Jin; Yuan Deng; Yichen Gong. Students' Perceptions of Active Learning Classrooms from an Informal Learning Perspective: Building a Full-Time Sustainable Learning Environment in Higher Education.

Sustainability, 14(14), 8578. DOI: [10.3390/su14148578](https://doi.org/10.3390/su14148578)

17. Lei Peng; Wenli Wei; Wenyi Fan; Shitao Jin; Yuxuan Liu. Student Experience and Satisfaction in Academic Libraries: A Comparative Study among Three Universities in Wuhan.

Buildings, 12(5), 682. DOI: [10.3390/buildings12050682](https://doi.org/10.3390/buildings12050682)

18. Lei Peng; Yuan Deng; Shitao Jin. The Evaluation of Active Learning Classrooms: Impact of Spatial Factors on Students' Learning Experience and Learning Engagement.

Sustainability, 14(8), 4839. DOI: [10.3390/su14084839](https://doi.org/10.3390/su14084839)

19. 彭雷; 范文奕; 靳世涛. 促进创新人才培养的大学校园学习环境要素分析：基于两所创新研究型大学校园空间规划的案例研究.

新建筑, 2022(6), 98-103. DOI: [10.12069/j.na.202206098](https://doi.org/10.12069/j.na.202206098)

Conference papers

1. Shitao Jin; Yuan Deng; Huijun Tu. Collective Intelligence in Architectural Programming: Multi-Agent Reinforcement Learning for Optimization in Mega Transportation Projects.

Proceedings of the 31st International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA), Vol. 1, 693-702.

2. Shitao Jin; Yuan Deng; Huijun Tu. Reprogramming Urban Communities: Enhancing Children's Well-being through a Value-Based Programming Approach in Inclusive Planning and Design. Proceedings of the 60th ISOCARP World Planning Congress, Siena, Italy. DOI: [10.47472/C6qUucJ8](https://doi.org/10.47472/C6qUucJ8)

TEACHING & COURSE DEVELOPMENT

- Architectural programming and design education: developed and evaluated an AI-assisted architectural programming/design course model published in Buildings (2024).
- Research-based teaching focus: AI tools for problem exploration, information gathering, decision-making, design iteration, and design evaluation.
- Teaching interests: architectural programming, AI for architectural design, construction/project management, research methods, and evidence-based design.

HONORS & AWARDS

- National Scholarship for PhD Students, 2024 and 2025
- National Scholarship for Master Students, 2022
- Outstanding Graduate; Three-Good Graduate; First-Class Academic Scholarship; Zhixing Excellence Award
- Jilin Provincial Government Scholarship; First-Class Undergraduate Academic Scholarship
- First Prize, National BIM Graduation Design Competition; Second Prize, National College English Competition

PEER REVIEW & ACADEMIC SERVICE

- Verified peer reviewer via Web of Science / Publons / ORCID: 146 reviews across 7 journals, 2024-2026.
- By year: 5 reviews in 2024, 56 in 2025, and 85 in 2026.
- Journal portfolio: Engineering, Construction and Architectural Management (73), Buildings (32), Sustainability (31), Systems (3), SAGE Open (3), Geocarto International (3), Journal of Asian Architecture and Building Engineering (1).
- Student leadership: class monitor during master's study; academic committee member and employment department deputy head during undergraduate study.

PROFESSIONAL PRACTICE

JUL 2022 - AUG 2022

Architectural Management Intern, China Railway Siyuan Survey and Design Group Co., Ltd.
Construction documentation for Nantong Station and Changzhou South Station; schematic design for Wenzhou East Station urban transport interchange center.

JUL 2021 - SEP 2021

Architectural Design Intern, China Architecture Design & Research Group
Green building design, energy-saving calculations, and construction drawings for Jinan Evergrande International Financial Center.

JUN 2019 - AUG 2019

Architectural Design Intern, Wuhan Hechuang Architectural Design Co., Ltd.
Campus planning and architectural design for Wuhan Optics Valley No. 13 Junior High School.

METHODS & TECHNICAL SKILLS

- Python, machine learning, natural language processing, deep learning, multi-agent reinforcement learning
- Bibliometrics, LDA topic modeling, sentiment analysis, social network analysis, entropy weight, fuzzy TOPSIS, Monte Carlo simulation
- Rhino, Grasshopper, Revit, AutoCAD, SketchUp, Photoshop, Illustrator, InDesign, Premiere Pro
- Stata, SPSS; IELTS 7.0

ACADEMIC REFERENCES / SUPERVISORS

Prof. Huijun Tu

College of Architecture and Urban Planning, Tongji University; doctoral supervisor; Director, Urban Architectural Programming and Evaluation Research Center; research areas: architectural design, urban design, architectural programming.

A/Prof Felix Kin Peng Hui

Department of Infrastructure Engineering, Faculty of Engineering and Information Technology, The University of Melbourne; Engineering Technology Management Expert.

A/Prof Paulo Vaz Serra

Faculty of Architecture, Building and Planning, The University of Melbourne; Associate Professor in Construction Management.